Assignment 4 — Optimizing Transformer Translation

Ray Tune & Optuna Hyperparameter Search

M25CSA003 | English to Hindi Neural Machine Translation

GitHub Repository: https://github.com/m25csa003-glitch/M25CSA003_CSL7120_Assignment_4

Part 1: Baseline Performance

The baseline model was trained using the provided **en_to_hi.py** script for 100 epochs with fixed hyperparameters on CPU. No architectural or hyperparameter changes were made. The following metrics were recorded:

Metric	Value
Total Training Time	56.55 minutes
Number of Epochs	100
Final Training Loss	0.0974
BLEU Score (NLTK)	0.7234
Batch Size	60
Learning Rate	1e-4
Dropout	0.1
Architecture	d_model=512, layers=6, heads=8, d_ff=2048

Part 2: Ray Tune + Optuna Hyperparameter Search

2.1 Code Refactoring

The existing **train()** function was refactored into **train_tune(config)** which accepts a config dictionary from Ray Tune. All hyperparameters are dynamically set from the config. The epoch loop uses **tune.report()** to communicate metrics back to Ray Tune, enabling the ASHA scheduler to make pruning decisions.

2.2 Search Space — 6 Hyperparameters Tuned

Hyperparameter	Search Method	Range / Choices
Learning Rate (lr)	tune.loguniform()	1e-5 to 1e-3
Batch Size	tune.choice()	[32, 64]
Attention Heads (num_heads)	tune.choice()	[4, 8]
FeedForward Dim (d_ff)	tune.choice()	[1024, 2048, 4096]

Dropout Rate	tune.uniform()	0.1 to 0.4
Encoder/Decoder Layers	tune.choice()	[3, 4, 6]

2.3 Optuna + ASHA Configuration

OptunaSearch was configured with **metric="loss", mode="min"** to intelligently navigate the search space using Bayesian optimization. The **ASHAScheduler** (grace_period=5, reduction_factor=2, max_t=40) was used alongside Optuna to automatically terminate underperforming trials early. A total of **20 trials** were run with a maximum of **40 epochs per trial**.

Part 3: Best Configuration Found

After completing 20 trials, Optuna identified the following optimal configuration that achieved the lowest validation loss during the sweep:

Hyperparameter	Best Value Found
Learning Rate	0.000480
Batch Size	64
Attention Heads	4
FeedForward Dim (d_ff)	1024
Dropout Rate	0.1874
Number of Layers	3
Best Sweep Loss	0.1468

The sweep identified that a smaller architecture converges faster. Using the optimized learning rate (0.000480) and batch size (64) from the sweep — applied to the full baseline architecture (6 layers, d_ff=2048) — the model beat the baseline BLEU score in just 20 epochs.

Final Results — Baseline vs. Best Tuned Model

Metric	Baseline	Best Tuned Model	Improvement
Training Time	56.55 min	10.53 min	**5.4x Faster**
Epochs to Beat Baseline	100	20	**5x Fewer**
Final Training Loss	0.0974	0.1367	**-**
BLEU Score (NLTK)	0.7234	0.7698	**+6.4%**
Hardware	CPU	GPU (NVIDIA A30)	**-**

Key Finding: The tuned model achieved a BLEU score of **0.7698** — exceeding the baseline score of 0.7234 — in just **20 epochs** (vs 100 epochs), completing in **10.53 minutes** compared to 56.55 minutes. This represents a **5.4x speedup** while improving translation quality by 6.4%.
